Schur-Stable Fixed-Anchor Certificates at Prescribed Degree: A Robust Converse and a Certified Finite Construction

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Technical Note 2, revised for discussion

October 21, 2026

Abstract

At a prescribed polynomial degree, robust separation of the convexified monomial dynamics is equivalent to a finite forward fixed-anchor vector barrier certificate with a constant nonnegative Schur-stable comparison matrix. This is exactly the class that can be shifted to have a uniform strict global propagation margin. It forces uniform eventual avoidance of the unsafe set from the entire verification domain, and therefore excludes onto maps and unsafe periodic orbits. A constructive strengthening gives an explicit component bound from a certified convex body, together with a finite linear program that accounts for source and successor approximation errors. A nonlinear worked certificate is replayed with exact rational arithmetic. The original same-degree recovery and moment obstruction are retained with their precise scopes; physical robust safety alone does not imply the feature-space hypothesis.

1 Setting and the convexified dynamics

Fix an integer $d \geq 1$. Assume $X \subseteq \mathbb{R}^n$ is nonempty and compact, $f : X \to X$ is continuous, and $X_0, X_u \subseteq X$ are nonempty, compact, and disjoint. All degree bounds below mean *total degree at most* d, with d chosen in advance. A forward fixed-anchor VBC consists of finitely many polynomial components $B = (B_1, \dots, B_m)^\top$, a constant entrywise nonnegative A, and $\delta_0, \delta_u > 0$ such that

$$B \leq -\delta_0 \mathbf{1} \text{ on } X_0, \qquad B_1 \geq \delta_u \text{ on } X_u, \qquad B(f(x)) \leq AB(x) \quad (x \in X). \tag{1}$$

The same designated component B_1 separates *all* unsafe states. No restriction on $\rho(A)$ is assumed. The converse in this note concerns the forward direction.

Let $\phi_d(x) \in \mathbb{R}^D$, $D = \binom{n+d}{d} - 1$, list the nonconstant monomials of degree at most d. Constants are represented by affine offsets, so every admissible polynomial is $B_i(x) = \beta_i(\phi_d(x))$ for an affine β_i. Define

$$\begin{aligned} &E = \operatorname{aff} \phi_d(X), \quad V = E - E, \quad Z = \operatorname{conv} \phi_d(X), \\ &I_d = \operatorname{conv} \phi_d(X_0), \qquad J_d = \operatorname{conv} \phi_d(X_u), \\ &\mathcal{F}_d = \operatorname{conv}\{(\phi_d(x), \phi_d(f(x))) : x \in X\} \subseteq E \times E. \end{aligned} \tag{2}$$

The convex sets Z, I_d, J_d, and $\mathcal{F}_d$ are compact; the affine space E and direction space V need not be bounded. The unit ball $\mathbb{B}_V$ is in V, and interiors, distances, and norms are relative to E. Omitting the constant feature avoids artificial perturbations of its identically-one coordinate.

A pair $(s, t) \in \mathcal{F}_d$ can be the feature averages of a distribution and its successor distribution, even when s and t are not individual state lifts. An affine comparison on the original graph holds on all of $\mathcal{F}_d$. Thus ordinary lifted trajectory safety does not suffice.

For $\varepsilon \geq 0$, put

$$\begin{aligned} \mathcal{F}_d^\varepsilon &= \mathcal{F}_d + (\varepsilon \mathbb{B}_V \times \varepsilon \mathbb{B}_V), \\ T_\varepsilon(C) &= \{t : \ \exists s \in C, \ (s, t) \in \mathcal{F}_d^\varepsilon\}, \\ C_\varepsilon^0 &= I_d + \varepsilon \mathbb{B}_V, \qquad C_\varepsilon^{k+1} = \operatorname{conv}\big(C_\varepsilon^k \cup T_\varepsilon(C_\varepsilon^k)\big), \\ \mathcal{K}_\varepsilon &= \operatorname{cl} \bigcup_{k \geq 0} C_\varepsilon^k. \end{aligned} \tag{3}$$

The source and successor errors are independent feature-space errors, not a claim about physical state disturbances. There are two convexifications: the transition graph in $\mathcal{F}_d$, and the reachable set at every iteration in (3). Each C_ε^k is compact and convex; the sets increase and lie in the compact set $Z + \varepsilon\mathbb{B}_V$. Consequently $\mathcal{K}_\varepsilon$ is compact and convex.

Every VBC satisfying (1) forces

$$\mathcal{K}_0 \cap J_d = \varnothing. \tag{4}$$

Indeed, $\{\beta \le 0\}$ contains I_d and is preserved by $\mathcal{F}_d$, convex combinations, and limits, whereas $\beta_1 \ge \delta_u$ on J_d. This is a necessary condition; no general converse to (4) is asserted here.

2 The converse and its exact comparison-matrix class

Theorem 1 (Robust convexified safety, strict propagation, and Schur stability)**.** *Under the assumptions above, the following are equivalent.*

1. *There is an $\varepsilon > 0$ such that $\mathcal{K}_\varepsilon \cap J_d = \varnothing$.*
2. *There are finite $m \ge 1$, polynomial B_i of degree at most d, a constant $A \ge 0$, and $\delta_0, \delta_u, \eta > 0$ satisfying the separation conditions in (1) and*
$$B(f(x)) \le AB(x) - \eta\mathbf{1} \qquad (x \in X). \tag{5}$$
3. *There is a VBC satisfying (1), at the prescribed degree bound, with a constant $A \ge 0$ satisfying $\rho(A) < 1$.*

Every matrix in item 2 is necessarily Schur stable. Items 2 and 3 can be interconverted without changing the degree or component count, although item 1 does not specify a minimal count. Moreover, the robust separation condition in item 1 yields a bounded polyhedron

$$P = \{z \in E : \beta_j(z) \le 0,\ j = 1, \dots, m\}$$

with I_d strictly inside it, one designated supporting inequality strictly separating J_d, and

$$(s,t) \in \mathcal{F}_d,\ s \in P \quad \Longrightarrow \quad t + \varepsilon\mathbb{B}_V \subseteq P.$$

Thus it yields a VBC of the prescribed degree bound in the usual class (1).

Scope. The equivalence characterizes the *Schur-stable subclass* of ordinary forward fixed-anchor VBCs, rather than all certificates in (1). Ordinary certificates can have $\rho(A) = 1$. The converse does not cover that boundary class. The existence proof below uses compactness and continuity; Sections 4–5 add finite information and explicit approximation guarantees.

2.1 Two finite-dimensional ingredients

Lemma 1 (Finite outer approximation with a prescribed separator)**.** *Let $K \subseteq E$ be a compact convex body, let $a_1 \in V$ have unit norm, and let $\tau > 0$. There are finitely many unit vectors $a_1, \dots, a_m \in V$ such that, with $h_K(a) = \max_{z \in K} \langle a, z\rangle$,*

$$K \subseteq P = \{z \in E : \langle a_j, z\rangle \le h_K(a_j),\ j = 1, \dots, m\} \subseteq K + \tau\mathbb{B}_V,$$

and P is bounded.

Proof. Choose an orthonormal basis of V and its positive and negative coordinate directions. Their supporting halfspaces bound a compact box Q containing K. Each $y \in Q$ with $\operatorname{dist}(y, K) \ge \tau$ is strictly excluded by a supporting halfspace: if p is its closest point in K, take $a = (y - p)/\|y - p\|$, so $\langle a, y\rangle > h_K(a) = \langle a, p\rangle$. The corresponding open exclusion sets cover the compact set $\{y \in Q : \operatorname{dist}(y, K) \ge \tau\}$. A finite subcover, the box inequalities, and the prescribed direction a_1 give the result. □

Lemma 2 (Global row multiplier from a convex relation)**.** *Let $F \subseteq E \times E$ be compact and convex, let $\beta_1, \dots, \beta_m$ be affine, and suppose some $(s_*, t_*) \in F$ has $\beta(s_*) < 0$. If*

$$p_i = \max\{\beta_i(t) : (s,t) \in F,\ \beta(s) \le 0\},$$

then there is an attained $a_i \in \mathbb{R}^m_{\geq 0}$ *such that*

$$\beta_i(t) - a_i^\top \beta(s) \leq p_i \quad \textit{for every } (s,t) \in F.$$

Proof. The closed convex set

$$H_i = \{(\beta(s) + u,\ \beta_i(t) - v) : (s,t) \in F,\ u \geq 0,\ v \geq 0\}$$

contains $(0, p_i)$ on its boundary. It is closed because it is a compact affine image plus a closed cone. A nonzero supporting normal (w, γ) can be oriented so that

$$w^\top y + \gamma q \leq \gamma p_i \quad ((y,q) \in H_i).$$

Then $w \leq 0$ and $\gamma \geq 0$, since u and v can increase without bound. If $\gamma = 0$, strict feasibility gives $w^\top \beta(s_*) > 0$ for nonzero $w \leq 0$, contradicting the displayed inequality. Hence $\gamma > 0$; dividing by γ and taking $a_i = -w/\gamma$ proves the claim. This is the usual affine-constraint convex duality argument. □

2.2 Proof of Theorem 1

Proof. (1)⇒(2): from the closure to finitely many facets. Write $K = \mathcal{K}_\varepsilon$. It has nonempty relative interior because it contains $I_d + \varepsilon \mathbb{B}_V$. Compact convex separation of K and J_d provides a unit $a_1 \in V$ and

$$\gamma = \min_{z \in J_d} \langle a_1, z \rangle - h_K(a_1) > 0. \tag{6}$$

Apply Lemma 1 with $\tau = \varepsilon/2$, and define

$$\beta_j(z) = \langle a_j, z \rangle - h_K(a_j), \qquad B_j(x) = \beta_j(\phi_d(x)).$$

Each polynomial has degree at most d. Since $I_d + \varepsilon \mathbb{B}_V \subseteq K$ and every a_j is a unit vector,

$$\beta_j(s) \leq -\varepsilon \quad (s \in I_d), \qquad \beta_1(z) \geq \gamma \quad (z \in J_d).$$

The key transition fact is

$$(s,t) \in \mathcal{F}_d,\ \operatorname{dist}(s,K) \leq \varepsilon/2 \quad \Longrightarrow \quad t + \varepsilon \mathbb{B}_V \subseteq K. \tag{7}$$

To prove it, choose $p \in K$ closest to s, and $p_\ell \in \bigcup_k C^k_\varepsilon$ tending to p. For large ℓ, $\|p_\ell - s\| < \varepsilon$. For every $w \in \varepsilon \mathbb{B}_V$, the pair $(p_\ell, t + w)$ belongs to $\mathcal{F}^\varepsilon_d$. If $p_\ell \in C^{k_\ell}_\varepsilon$, then $t + w \in C^{k_\ell + 1}_\varepsilon \subseteq K$. This proves (7). In particular, the proof does not assume that passing to the closure automatically preserves every boundary source fiber of the perturbed relation.

For $(s,t) \in \mathcal{F}_d$ with $s \in P$, the outer approximation gives $\operatorname{dist}(s,K) \leq \varepsilon/2$. By (7), $t + \varepsilon \mathbb{B}_V \subseteq K$, hence

$$\beta_i(t) \leq -\varepsilon \quad (i = 1, \ldots, m).$$

There is a strictly feasible source in $\mathcal{F}_d$, obtained from any $x_0 \in X_0$. Lemma 2, applied row by row, therefore supplies $A_i \geq 0$ with

$$\beta_i(t) - A_i \beta(s) \leq -\varepsilon \quad ((s,t) \in \mathcal{F}_d).$$

Stacking the rows and restricting to the original graph yields (5) with $\eta = \delta_0 = \varepsilon$ and $\delta_u = \gamma$. The same argument gives the stated relational invariance of P.

(2)⇒(1): strict propagation tolerates feature errors. Write the linear part of β_i on V as a_i, let $L_i = \|a_i\|$, and set

$$L = \max_i L_i > 0, \qquad M = \max_i \left(L_i + \sum_j A_{ij} L_j \right) > 0.$$

The positivity follows from the separating anchor. Choose

$$0 < \varepsilon < \min\{\delta_0/(2L),\ \eta/(2M)\}.$$

Affine linearity extends (5) from the graph to $\mathcal{F}_d$. For $(s,t) \in \mathcal{F}_d$ and $e, w \in \varepsilon \mathbb{B}_V$,

$$\begin{aligned}\beta_i(t+w) &\le (A\beta(s+e))_i - \eta + \varepsilon \left(L_i + \sum_j A_{ij} L_j \right) \\ &\le (A\beta(s+e))_i - \eta/2.\end{aligned}$$

Also $\beta \le -\delta_0 \mathbf{1}/2$ on C_ε^0. Nonnegativity of A and the displayed perturbed comparison imply inductively $\beta \le 0$ on every C_ε^k; affine inequalities persist under convex combinations and limits. Thus $\beta \le 0$ on $\mathcal{K}_\varepsilon$, whereas $\beta_1 \ge \delta_u$ on J_d. The sets are disjoint.

(2)⇒(3): strict propagation forces Schur stability. For a nonzero left Perron vector $w \ge 0$, write $w^\top A = \rho(A) w^\top$. Fix $x_0 \in X_0$ and $q_k = w^\top B(f^k(x_0))$. Then $q_0 < 0$ and

$$q_{k+1} \le \rho(A) q_k - \eta w^\top \mathbf{1}.$$

If $\rho(A) \ge 1$, induction makes this upper bound tend to $-\infty$, contradicting boundedness of B on the compact forward-invariant X. Thus $\rho(A) < 1$. The same argument without η excludes $\rho(A) > 1$ for every ordinary VBC in (1); consequently such a VBC always has $\rho(A) \le 1$.

(3)⇒(2): a constant shift supplies strictness. Let $v = (I-A)^{-1}\mathbf{1} = \sum_{j \ge 0} A^j \mathbf{1} \ge \mathbf{1}$. Choose $0 < c < \delta_u / v_1$ and put $\widetilde{B} = B - cv$. Since $v - Av = \mathbf{1}$,

$$\widetilde{B} \circ f \le A\widetilde{B} - c\mathbf{1}, \qquad \widetilde{B} \le -\delta_0 \mathbf{1} - cv \text{ on } X_0, \qquad \widetilde{B}_1 \ge \delta_u - cv_1 > 0 \text{ on } X_u.$$

This preserves the degree and number of components. □

Remark (Rational approximation). Strict margins on compact X also imply existence of a VBC with rational polynomial coefficients and a rational matrix $A \ge 0$, at the same degree and component count. Indeed, the separation values and the residual $AB - B \circ f$ depend continuously on these finitely many coefficients, uniformly on X, since $f(X) \subseteq X$. Sufficiently close rational approximations, keeping $A \ge 0$ and errors below half the respective positive margins, preserve all inequalities. This is an existence guarantee, not a computable rounding procedure for unspecified compact data.

Corollary 1 (A sufficient condition that can be checked on a chosen body)**.** *It suffices to exhibit a compact convex $C \subseteq E$ and $\varepsilon > 0$ satisfying*

$$I_d + \varepsilon \mathbb{B}_V \subseteq C, \qquad T_\varepsilon(C) \subseteq C, \qquad C \cap J_d = \varnothing.$$

Then $\mathcal{K}_\varepsilon \subseteq C$, and Theorem 1 constructs a finite degree-d fixed-anchor VBC.

This is a robust convex-body assumption, not an assumption that a finite polyhedral invariant already exists. The theorem uses supporting halfspaces to obtain finitely many functions and the global comparison matrix.

Remark (Finite continuous features). The proof uses only compactness of the graph image and affinity of the component functions in feature space. It therefore applies to any fixed finite continuous feature map, with affine offsets representing constants and with all balls relative to its affine hull. The monomial choice supplies the prescribed-polynomial-degree interpretation. No computational claim follows from continuity alone.

3 Dynamical consequences and scope limitations

Proposition 1 (Uniform eventual avoidance on the entire domain)**.** *Every strict certificate in Theorem 1 has a common integer N such that $B(f^k(x)) < 0$ componentwise for every $x \in X$ and $k \ge N$. Thus all trajectories in the entire verification domain eventually avoid X_u, uniformly in their starting point. The same avoidance conclusion holds for ordinary Schur-stable certificates, using their shifted functions.*

Proof. Iteration gives

$$B(f^k(x)) \le A^k B(x) - \eta \sum_{j=0}^{k-1} A^j \mathbf{1} = A^k (B(x) + \eta v) - \eta v, \quad v = (I-A)^{-1}\mathbf{1}.$$

Since $\rho(A) < 1$ and B is bounded, the right side tends uniformly to $-\eta v < 0$. More quantitatively, let

$$\kappa = \max_i \frac{(Av)_i}{v_i} = \max_i (1 - v_i^{-1}) < 1, \qquad H = \max\left\{0, \max_{x \in X, i} \frac{B_i(x) + \eta v_i}{v_i}\right\}.$$

Nonnegativity of A yields $B(f^k(x)) \le (H\kappa^k - \eta)v$. Any N with $H\kappa^N \le \eta/2$ suffices. If $\kappa = 0$, take $N = 1$; if $H = 0$, take $N = 0$. For an ordinary Schur-stable certificate apply this argument to $\widetilde{B}$ from the theorem; the original unshifted functions need not become negative. □

Corollary 2 (Obstructions at every degree)**.** *An unsafe periodic point excludes strict and ordinary Schur-stable fixed-anchor VBCs at every degree and finite component count. If f is onto X, these classes are also excluded whenever $X_u \neq \varnothing$.*

Proof. An unsafe periodic point is visited at arbitrarily late times, contradicting Proposition 1. If f is onto, $f^N(X) = X$; uniform eventual negativity would force the anchor to be negative on all of X, including X_u. □

These are limitations of the additional strict/Schur-stable class, not obstructions to all ordinary certificates.

4 A quantitative finite-facet construction

We now strengthen the chosen-body criterion (Corollary 1). The body is an input to be certified, not an unknown infinite-horizon closure whose support values are assumed available. Fix $z_c \in E$, let $r = \dim V \ge 1$, and write

$$h_C(a) = \max_{z \in C} \langle a, z - z_c \rangle.$$

Assume a compact convex body C and certified constants $\varepsilon, \gamma, R > 0$, with $R \ge \varepsilon$, satisfy

$$I_d + \varepsilon \mathbb{B}_V \subseteq C, \quad T_\varepsilon(C) \subseteq C, \quad C \subseteq z_c + R\mathbb{B}_V, \quad \min_{z \in J_d} \langle a_1, z - z_c \rangle \ge h_C(a_1) + \gamma, \tag{8}$$

where $a_1 \in V$ is a certified unit anchor direction.

Lemma 3 (A coarse explicit support net)**.** *Set $\delta = \varepsilon/(16R)$ and choose $0 \le \alpha \le \min\{\varepsilon/8, \gamma/2\}$. There is a finite δ-net Q of the unit sphere in V, in Euclidean chord distance, with*

$$|Q| \le \lceil (1 + 2/\delta)^r - (2/\delta - 1)^r \rceil.$$

Use the directions $Q \cup \{a_1\}$, with the anchor first, and certified support upper bounds

$$h_C(a_i) \le \widehat{h}_i \le h_C(a_i) + \alpha.$$

For $\beta_i(z) = \langle a_i, z - z_c \rangle - \widehat{h}_i$ and $P = \{\beta \le 0\}$,

$$C \subseteq P \subseteq C + \tfrac{\varepsilon}{2}\mathbb{B}_V.$$

Proof. Choose a maximal δ-separated subset of the sphere. The disjoint open balls of radius $\delta/2$ about its points lie in the annulus with radii $1 - \delta/2$ and $1 + \delta/2$. Comparing r-dimensional volumes proves the cardinality bound and ensures the selection is finite. Maximality gives the net property.

For $z \in P$, let $u = (z - z_c)/\|z - z_c\|$ when $z \neq z_c$ and take $q \in Q$ with $\|q - u\| \le \delta$. As $\langle q, u \rangle \ge 1 - \delta$, we have

$$\|z - z_c\| \le (R + \alpha)/(1 - \delta).$$

For every unit a take a net direction q within δ. Since the support function is R-Lipschitz on the unit sphere,

$$\begin{aligned} \langle a, z - z_c \rangle - h_C(a) &\le \alpha + \delta\|z - z_c\| + R\delta \\ &\le 2\alpha + 4R\delta \le \varepsilon/2. \end{aligned}$$

Here $\delta \le 1/16$, so $(1 - \delta)^{-1} \le 16/15$ suffices for the second bound. The support-halfspace description of $C + (\varepsilon/2)\mathbb{B}_V$ proves the inclusion. Lower support bounds give $C \subseteq P$. □

Theorem 2 (Explicit component bound from a certified body)**.** *Under* (8) *and the support-value assumptions of Lemma 3, a strict forward fixed-anchor VBC exists with*

$$m \le 1 + \lceil (1 + 32R/\varepsilon)^r - (32R/\varepsilon - 1)^r \rceil, \tag{9}$$

and margins $\delta_0 \ge \varepsilon$, $\delta_u \ge \gamma - \alpha \ge \gamma/2$, *and* $\eta \ge \varepsilon$. *Its matrix is necessarily Schur stable. The geometric count is* $O_r((R/\varepsilon)^{r-1})$*; no optimality is claimed.*

Proof. Use Lemma 3. The initial tube gives $\beta_i \le -\varepsilon$ on I_d, and the anchor gives the stated unsafe margin. For a nominal pair $(s,t) \in \mathcal{F}_d$ with $\operatorname{dist}(s,C) \le \varepsilon$, choose a closest $y \in C$. Then

$$(y, t + w) \in \mathcal{F}_d^{\varepsilon} \quad (w \in \varepsilon \mathbb{B}_V), \qquad t + \varepsilon \mathbb{B}_V \subseteq C.$$

This uses the certified inclusion $T_\varepsilon(C) \subseteq C$ directly. In particular it holds for $s \in P$, so $\beta_i(t) \le -\varepsilon$. Lemma 2 supplies the global nonnegative comparison rows. Theorem 1 gives Schur stability. The bound counts the net and at most one additional anchor direction. □

The unsafe margin γ controls the required support accuracy through α. It does not enlarge the count because its separating direction is supplied explicitly.

The same row argument works for *any* certified unit-normal outer polyhedron satisfying

$$C \subseteq P \subseteq C + \varepsilon \mathbb{B}_V, \tag{10}$$

with the stated support errors and anchor. Thus a more efficient polyhedron can replace the coarse universal net. Supporting-halfspace approximation and preservation of invariance within a dynamical margin are classical mechanisms; the count here is a quantitative consequence for this degree-restricted certificate class.

5 Recovering the matrix from finite information

Assume also that $Z \subseteq z_c + R_Z \mathbb{B}_V$ is certified. Put

$$D = R_Z + R + \alpha, \qquad \Lambda = D/\varepsilon.$$

For the facets just constructed, the row multipliers with nominal margin ε can be chosen to have row sums at most Λ. In fact, at any initial graph pair (s_*, t_*) we have $\beta_j(s_*) \le -\varepsilon$ and $\beta_i(t_*) \ge -D$. Hence every such row satisfies

$$-D + \varepsilon \sum_j A_{ij} \le \beta_i(t_*) - \sum_j A_{ij} \beta_j(s_*) \le -\varepsilon, \qquad \sum_j A_{ij} \le (D - \varepsilon)/\varepsilon < \Lambda.$$

A smaller row cap is permitted if a separate bound proves that nominal dual rows with margin ε satisfy it.

Let $(s_\ell, t_\ell) = (\phi_d(x_\ell), \phi_d(f(x_\ell)))$ be *exact graph pairs* at finitely many $x_\ell \in X$. Suppose a certified paired cover gives, for every $x \in X$, a common index ℓ with

$$\|\phi_d(x) - s_\ell\| \le \xi_s, \qquad \|\phi_d(f(x)) - t_\ell\| \le \xi_t. \tag{11}$$

These are errors in the original features, before applying the facets. Choose a target propagation margin $\eta_* > 0$ satisfying

$$\xi_t + \Lambda \xi_s + \eta_* \le \varepsilon. \tag{12}$$

For each row i, solve the following *finite linear feasibility problem*:

$$\begin{aligned} A_{ij} \ge 0, \qquad \sum_j A_{ij} \le \Lambda, \\ \beta_i(t_\ell) - \sum_j A_{ij} \beta_j(s_\ell) + \xi_t + \xi_s \sum_j A_{ij} \le -\eta_* \qquad \text{for every } \ell. \end{aligned} \tag{13}$$

Theorem 3 (Feasible finite LP with continuum soundness)**.** *Under the certified body, facet, row-cap, and exact paired-cover assumptions above,* (13) *is feasible. Every feasible solution gives*

$$B \circ f \le AB - \eta_* \mathbf{1} \quad \text{on all of } X, \qquad B_i = \beta_i \circ \phi_d,$$

retaining the certified initial and unsafe margins. This inequality also holds on the full convexified relation $\mathcal{F}_d$.

Proof. Nominal dual rows with margin ε satisfy every exact node constraint with residual at most $-\varepsilon$. Their row caps and (12) prove LP feasibility. For any returned row and any x, choose the common node in (11). Unit facet normals and $A_{ij} \geq 0$ bound the change in the residual by $\xi_t + \xi_s \sum_j A_{ij}$. The node constraint therefore implies the desired inequality at x. Affinity in the paired features extends it to convex combinations, and continuity to limits. Initial and unsafe separation were already certified geometrically. □

This LP does not impose a nonconvex spectral constraint: strict propagation and initial separation imply $\rho(A) < 1$ by Theorem 1. Exact rational LP data admit a rational feasible solution. Floating-point discovery requires a separate exact or interval verification of (13); a solver's success status alone is insufficient. If node evaluations themselves are approximate, their certified errors must also be charged to the budget. The feasibility proof as written assumes exact graph nodes.

Effective inputs and termination. A conditional construction takes as input a certificate of (8), geometry bounds, certified support-value oracles, and a finite paired-cover oracle. A maximal net takes finitely many selections by Lemma 3; finitely many support calls and finitely many LPs then finish the construction. The procedure does not discover C automatically.

One explicit admissible representation is compact semialgebraic X, X_0, X_u and C, polynomial f, and algebraic coefficients, with certified algebraic feature coordinates and bounds. Real-algebraic decision/optimization oracles can verify the body conditions, separator, and support bounds. By Carathéodory's theorem, a pair in $\mathcal{F}_d \subseteq E \times E$ needs at most $2r + 1$ graph atoms, so its use in the transition-inclusion formula is finite. Sphere-net selection is also a semialgebraic feasibility query. On a known bounding box, certified Lipschitz bounds L_s for ϕ_d and L_t for $\phi_d \circ f$ give (11) from a state h-net with $\xi_s = L_s h$, $\xi_t = L_t h$. A finite box subdivision, exact nonempty-cell decisions, and one exact point of X per occupied cell yield such a cover. If $L_t + \Lambda L_s > 0$, choosing

$$h \leq \frac{\varepsilon - \eta_*}{L_t + \Lambda L_s}$$

suffices. The zero-denominator case needs only exact constant feature pairs. Algebraic LPs may be solved with exact algebraic arithmetic; rational data allow rational LPs. These statements give finite termination with the stated oracles, not efficiency or a general compact-continuous algorithm. In the worked example all body and cover conditions have direct analytic certificates and all LP checks are rational.

6 A nonlinear positive example with full feature errors

Take

$$X = [-1, 1], \quad f(x) = \tfrac{1}{2}x + \tfrac{1}{4}x^2, \quad X_0 = [-\tfrac{1}{4}, \tfrac{1}{4}], \quad X_u = [\tfrac{7}{10}, \tfrac{4}{5}], \quad d = 2.$$

Since $f'(x) = (1 + x)/2 \in [0, 1]$, $f(X) = [-1/4, 3/4] \subseteq X$. Use $\phi_2(x) = (x, x^2)$ and $z_c = 0$. For *every* pair $(s, t) \in \mathcal{F}_2$,

$$t_1 = \tfrac{1}{2}s_1 + \tfrac{1}{4}s_2, \quad 0 \leq t_2 \leq \tfrac{9}{16}s_2, \quad s_2 \geq 0.$$

These follow by averaging the graph equalities and the pointwise bound $f(x)^2 = x^2(1/2 + x/4)^2 \leq (9/16)x^2$. Consequently,

$$\|t\|_2^2 \leq \left(\tfrac{1}{4} + \tfrac{1}{16} + \tfrac{81}{256}\right) \|s\|_2^2 = \tfrac{161}{256}\|s\|_2^2 \leq (\tfrac{13}{16})^2 \|s\|_2^2.$$

This is a bound on mixed transition pairs, not just individual trajectories.

Let $C = (3/5)\mathbb{B}_2$, $\varepsilon = 1/20$, and $\gamma = 1/10$, with anchor $a_1 = (1, 0)$. On I_2, $\|s\|_2 \leq 5/16$, so $I_2 + \varepsilon\mathbb{B}_2 \subseteq C$. For any perturbed source $y = s + e \in C$ and independent successor error w, with $\|e\|_2, \|w\|_2 \leq \varepsilon$,

$$\|t + w\|_2 \leq \tfrac{13}{16}(3/5 + \varepsilon) + \varepsilon = \tfrac{37}{64} < \tfrac{3}{5},$$

leaving the certified inclusion gap $7/320$. The anchor strictly separates J_2 by γ. Thus the full robust convexified hypothesis holds. Notice that $f(1) = 3/4 \in X_u$; global avoidance in a single step is not assumed.

Theorem 2 gives the coarse universal bound $m \leq 1537$ ($r = 2$, $R/\varepsilon = 12$). A smaller certified construction uses twelve rational unit normals, all with exact offset $3/5$:

$$a_1 = (1, 0), \ a_2 = (0, 1), \ a_3 = (-1, 0), \ a_4 = (0, -1),$$
$$a_5 = (3/5, 4/5), \ a_6 = (3/5, -4/5), \ a_7 = (-3/5, 4/5), \ a_8 = (-3/5, -4/5),$$
$$a_9 = (4/5, 3/5), \ a_{10} = (4/5, -3/5), \ a_{11} = (-4/5, 3/5), \ a_{12} = (-4/5, -3/5).$$

For $P = \{a_i^\top z \le 3/5\}$, exact two-facet vertex enumeration gives $\max_{z\in P} \|z\|_2^2 = 2/5 < (13/20)^2$, so (10) holds. This is a direct geometric certificate; these twelve normals are not claimed to be the much finer universal net.

At the graph pair $(0,0)$, all facets equal $-3/5$, so nominal dual rows with margin ε have row sums at most $11/12$. We may therefore take the sharper LP cap $\Lambda = 1$. Use the 161 exact nodes $x_\ell = -1 + \ell/80$, $\ell = 0,\dots,160$. Their covering radius is $h = 1/160$. The derivative estimates

$$\|\phi_2'(x)\|_2^2 = 1 + 4x^2 \le 5 < 3^2, \qquad \|(\phi_2 \circ f)'(x)\|_2^2 = f'(x)^2(1 + 4f(x)^2) \le 13/4 < 2^2$$

give $\xi_s = 3/160$, $\xi_t = 1/80$, and $\xi_t + \Lambda\xi_s = 1/32$. LP discovery used target $1/80$; exact replay accepts the target $\eta_* = 1/100$ with minimum LP slack $1/400$.

Specifically, for $B_i(x) = a_{i1}x + a_{i2}x^2 - 3/5$, take the matrix with only

$$A_{1,5} = 28/125, \qquad A_{5,5} = 52/125, \qquad A_{9,5} = 58/125$$

nonzero. The finite constraints (13) hold exactly at all 161 nodes. Theorem 3 certifies the global continuum inequalities, including every convexified transition pair, with

$$\delta_0 \ge \tfrac{23}{80}, \qquad \delta_u \ge \tfrac{1}{10}, \qquad \eta \ge \tfrac{1}{100}, \qquad \rho(A) \le \|A\|_\infty = \tfrac{58}{125} < 1.$$

This example verifies an end-to-end finite construction. It does not claim that twelve components or degree two are necessary.

7 Adding functions at the same degree succeeds

Consider a contracting quarter-turn:

$$X = \{x \in \mathbb{R}^2 : \|x\|_2 \le 1\}, \quad f(x) = qRx, \quad q = \tfrac{9}{10}, \quad R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},$$

$$X_0 = \{x : \|x\|_2 \le \tfrac{1}{4}\}, \qquad X_u = \{(\tfrac{3}{4}, 0)\}, \qquad d = 1.$$

The system maps X into itself, and initial trajectories contract.

One affine function is impossible. Suppose $B(x) = a^\top x - c$ is a scalar VBC with multiplier $\lambda \ge 0$. Strict initial separation at 0 gives $c > 0$, while the unsafe anchor gives

$$c < \tfrac{3}{4}a_1 \le \tfrac{3}{4}\|a\|_2 < q\|a\|_2, \qquad a \ne 0.$$

At $x_* = R^\top a/\|a\|_2 \in X$, skew symmetry gives $a^\top x_* = 0$. Consequently

$$B(x_*) = -c < 0, \qquad B(f(x_*)) = q\|a\|_2 - c > 0,$$

contradicting $B(f(x_*)) \le \lambda B(x_*)$.

Four affine functions succeed. Take $r = 1/2$ and

$$B(x) = \begin{pmatrix} x_1 - r \\ x_2 - r \\ -x_1 - r \\ -x_2 - r \end{pmatrix}, \qquad A = q \begin{pmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

Direct substitution gives the global identity

$$B(f(x)) = AB(x) - \tfrac{1}{20}\mathbf{1}, \qquad \rho(A) = 9/10.$$

All components are at most $-1/4$ on X_0, and $B_1(3/4, 0) = 1/4$. The associated invariant is the square $[-1/2, 1/2]^2$. This proves a one-versus-four recovery at the same degree, without claiming that four is the minimum possible number.

The robust hypothesis also has a direct check. Since $\phi_1(x) = x$ and f is linear, $\mathcal{F}_1$ is its graph over the unit disk. Set $C = [-r, r]^2$ and $\varepsilon = 1/100$. Then $I_1 + \varepsilon\mathbb{B}_V \subseteq C$. For a perturbed source $y = s + e \in C$ and successor $qRs + w$, with $\|e\|_2, \|w\|_2 \le \varepsilon$,

$$\|qRs + w\|_\infty \le qr + (q+1)\varepsilon = \tfrac{469}{1000} < \tfrac{1}{2}.$$

Thus $T_\varepsilon(C) \subseteq C$, and C excludes J_1. Corollary 1 applies. This calculation includes every convexified transition pair and every subsequent convex combination.

8 An obstruction: adding same-degree functions cannot succeed

Reuse the nonlinear example from the earlier note:

$$f(x) = \tfrac{7}{8}x + \tfrac{5}{8}x^3 - \tfrac{1}{2}x^5, \qquad X = [-1, 1],$$
$$a = \tfrac{12}{25}, \quad u = \tfrac{1641}{3200}, \quad v = \tfrac{103}{200},$$
$$X_0 = [-a, a], \qquad X_u = [-v, -u] \cup [u, v].$$

Writing $z = x^2$ gives

$$f'(x) = \tfrac{1}{4} + (1 - z)(\tfrac{5}{8} + \tfrac{5}{2}z) \geq \tfrac{1}{4} \quad (x \in X);$$

together with $f(\pm 1) = \pm 1$, this proves that f is *onto* X. There is no true unsafe trajectory from X_0: the single quadratic

$$b(x) = x^2 - \tfrac{1}{4}$$

is an *interpolation-inspired barrier certificate (IBC)* with one frame. If $b(x) \leq 0$, then $f(x) = xh(x^2)$, where $h(z) = 1 + \frac{1}{2}(z - \frac{1}{4})(1 - z)$ lies in $[7/8, 1]$ for $0 \leq z \leq 1/4$. Hence $|f(x)| \leq |x| \leq 1/2$. The initial and unsafe margins are positive because $a < 1/2 < u$.

Nevertheless, at degree two put

$$\mu = \tfrac{369}{625}\delta_0 + \tfrac{128}{625}(\delta_{-3/4} + \delta_{3/4}), \qquad \mu_0 = \tfrac{1}{2}(\delta_{-a} + \delta_a), \qquad \nu = \tfrac{1}{2}(\delta_{-u} + \delta_u).$$

All are probability measures with support in their designated sets. Since $f(3/4) = 1641/2048$ and $u = af(3/4)/(3/4)$,

$$\int \phi_2 \, \mathrm{d}\mu = \int \phi_2 \, \mathrm{d}\mu_0 = (0, a^2), \qquad \int \phi_2 \circ f \, \mathrm{d}\mu = \int \phi_2 \, \mathrm{d}\nu = (0, u^2). \tag{14}$$

Thus $\big((0, a^2), (0, u^2)\big) \in \mathcal{F}_2$ has source in I_2 and successor in J_2. Already C_0^1 meets J_2, and so does $\mathcal{K}_\varepsilon$ for every $\varepsilon \geq 0$.

Proposition 2 (No finite same-degree repair)**.** *For this example no forward fixed-anchor VBC of degree at most two exists, with any finite component count and any constant $A \geq 0$. The conclusion also holds through degree three.*

Proof. For any degree-at-most-two vector B, (14) and the unit masses imply

$$\int B \circ f \, \mathrm{d}\mu = \int B \, \mathrm{d}\nu \leq A \int B \, \mathrm{d}\mu = A \int B \, \mathrm{d}\mu_0 \leq 0.$$

The designated unsafe component has integral at least $\delta_u > 0$, a contradiction. All odd moments through degree three vanish by symmetry, so the same identities and argument apply to cubic components. □

The source atoms at $\pm 3/4$ lie outside X_0. Their mixtures imitate initial features while the successor mixtures imitate unsafe features. This is precisely the information that ordinary trajectory safety omits. The fixed-anchor restriction is essential: positivity of some component at each unsafe point does not in general imply positivity of one component averaged over ν.

The quartic recovery belongs to the neutral boundary class. The earlier note gives the ordinary scalar quartic

$$p(x) = \tfrac{3}{10000} - (x^2 - \tfrac{1}{4})^2, \qquad A = [1],$$

with strict initial/unsafe separation and $p \circ f \leq p$ on X. Its exact verification is retained in the repository. Since this f is onto, Corollary 2 excludes *every* strict or Schur-stable VBC at every degree and component count. Thus the robust converse does not explain the quartic recovery. Every ordinary VBC here must have $\rho(A) = 1$: the initial-set Perron argument gives $\rho(A) \leq 1$, and onto-ness excludes $\rho(A) < 1$.

9 Physical robust safety does not imply feature robustness

Consider the nonidentity map

$$X = [-1, 1], \quad f(x) = x^2, \quad X_0 = [-1/4, 1/4], \quad X_u = [3/4, 1].$$

The ordinary certificate $B(x) = x^2 - 1/2$, $A = [1]$ has

$$B \le -7/16 \text{ on } X_0, \qquad B \ge 1/16 \text{ on } X_u, \qquad B(f(x)) - B(x) = x^4 - x^2 \le 0 \text{ on } X.$$

Yet $1 \in X_u$ is a fixed point. No strict or Schur-stable fixed-anchor VBC exists at any degree or finite component count. By Theorem 1, the robust convexified-feature hypothesis fails at every prescribed degree.

In contrast, additive *state* disturbances $|d_k| \le 1/8$ preserve the safe interval $C_{\text{state}} = [-1/2, 1/2]$, because

$$f(C_{\text{state}}) + [-1/8, 1/8] = [-1/8, 3/8] \subseteq C_{\text{state}}.$$

This interval contains X_0 and avoids X_u, proving physical robust safety. Independent source and successor feature errors can move off the realizable monomial variety; they encode a different and stronger requirement in this example.

The original identity example gives the same ordinary-versus-strict distinction at degree one: $f(x) = x$ on $X = [-1, 1]$, $X_0 = [-1/2, 0]$, $X_u = [1/2, 1]$, and $B(x) = x - 1/4$, $A = [1]$ satisfy ordinary separation and propagation. Every $\varepsilon > 0$ allows successive feature increments of size at most ε from 0 to $1/2$, so $\mathcal{K}_\varepsilon$ reaches J_1.

10 Comparison with existing results

Horváth–Song–Terlaky, Theorem 3.10. For a prescribed convex sublevel set $\{g \le 0\}$, [1] assumes convex g, concave $g \circ f_d$, and strict feasibility, and characterizes invariance by a global scalar multiplier:

$$g(f_d(x)) \le \lambda g(x) \quad (x \in \mathbb{R}^n), \qquad \lambda \ge 0.$$

If both functions are quadratic, the S-lemma removes the successor-concavity requirement; Remark 3.11 treats multiple inequalities. Lemma 2 uses the same classical convex-duality mechanism on a compact convex relation. The distinction to assess is construction of a finite family at the prescribed degree from robust convexified separation, together with its exact Schur-stable scope. It is not a new multiplier principle.

Liu, Section IV and Theorem 30. The discrete-time result in [2] assumes robust safety under additive state disturbances, separation of the closure of the disturbed reachable set from the unsafe set, local Lipschitz regularity, and compactness of that closure or global Lipschitz continuity. It yields a smooth robust barrier for a smaller disturbance radius. It does not prescribe polynomial degree or a constant nonnegative comparison matrix. Section 9 proves that its physical-robustness premise alone need not imply our feature-robustness premise; neither a stronger global margin nor a fixed degree follows merely from physical safety.

Alessio–Lazar–Bemporad–Heemels, Theorem 4. Theorem 4 in [3] is a general set-inclusion result for perturbed nonlinear discrete-time dynamics: fitting an intermediate set within a contraction margin preserves an appropriate invariance/contraction property. The theorem itself is not restricted to ellipsoids. Ellipsoids and piecewise ellipsoids enter their subsequent finite polyhedral construction. Their uncertainty model already allows multiple successors. Consequently neither the approximation-within-margin mechanism nor set-valuedness is an independent novelty here.

Finite polyhedral synthesis more broadly. Berger–Sankaranarayanan [4], Sections 6–7, gives a terminating counterexample-guided LP procedure with a quantitative iteration bound for polyhedral Lyapunov functions of continuous-time piecewise linear systems. Thus finite LP synthesis and robustness-dependent complexity bounds also have close precedents. Our conditional construction concerns a compact nonlinear discrete-time graph, fixed polynomial features, unsafe-anchor separation, and a global comparison matrix. These differences specify a comparison to make; they do not by themselves establish a publication-level novelty claim.

Issue	Earlier mechanisms	Result developed here
Invariance comparison	Global multipliers from convex duality	Same tool, applied after full graph convexification
Finite approximation	Fit within a dynamical margin	Same mechanism; explicit count for chosen feature body
Converse scope	Smooth robust barriers or given invariant sets	Finite degree-bounded fixed-anchor VBC iff feature robustness, exactly $\rho(A) < 1$
Finite construction	Existing geometric and LP methods	Certified paired feature cover and feasible row LP, with two error budgets

11 Assessment, remaining questions, and reproducibility

The Schur characterization, dynamical limitations, and physical-robustness counterexample formalize observations supplied in Majid Zamani's comments. The quantitative support bound and certified finite row-LP construction are the additional steps developed in this revision.

The revision proves the spectral characterization requested in the comments, makes its stronger dynamical limitations explicit, and adds two constructive consequences: the bound (9) and the finite feasible LP (13). The nonlinear example verifies their prerequisites on the entire convexified relation and supplies rational continuum margins. The general theorems are mathematical proofs in this note; their formal machine verification is not claimed.

The remaining research questions are substantive. The ordinary boundary class $\rho(A) = 1$ still lacks a general converse here; onto examples show it cannot be obtained by silently strengthening to uniform strictness. Discovery of a suitable body C is not solved by the conditional construction. The component count is coarse and no optimal count is asserted. Finally, novelty of the combined feature-space equivalence and constructive extraction must be established against the closest invariant-approximation and converse-certificate literature before a strong publication claim is warranted. The current results support a precise research direction; a publication-level novelty claim remains to be established.

Code and reproducibility. All computational materials are in the public Certificate Expressiveness repository:

`https://github.com/rezairaji60/certificate-expressiveness`

The Overleaf-ready source is `paper/converse.tex`; the exact nonlinear matrix and its deterministic report are `evidence/nonlinear_comparison_matrix.json` and `evidence/constructive_converse_report.json`. Run `python src/constructive_converse.py` for a standard-library-only exact replay. Optional `-discover` uses SciPy solely to propose a matrix, which is accepted only after rational verification. The existing `src/exact_verify.py` and `src/exact_verify_converse.py` retain the original continuum, rotation, and moment checks. Tests reject corrupted comparison matrices and understated feature-cover errors as well as invalid moment witnesses. The finite cover's soundness rests on the stated analytic derivative bounds; finite node checks without those bounds would not certify the continuum.

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